

Hybrid recommender systems

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Slides by: Christina Lioma

Lecture outline

- Main types of hybrid recommender systems
 - Output combination

Main recommender models

Approach	Input
Collaborative Filtering	My ratings + others' ratings
Content-based	My ratings + item features
Knowledge-based	My specification + item features + domain knowledge

Main recommender models

Approach	Input
Collaborative Filtering	My ratings + others' ratings
Content-based	My ratings + item features
Knowledge-based	My specification + item features + domain knowledge
Hybrid	Combinations of the above

Hybrid recommendation

Three main types

- Ensemble
- Monolithic
- Mixed

Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
- Monolithic
- Mixed

Hybrid recommendation

Three main types

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HYBRID OUTPUT

Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
- Monolithic: create a new recommender model
- Mixed

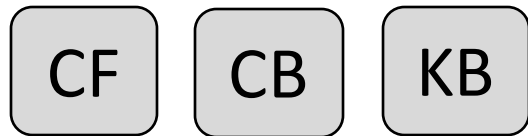


HYBRID OUTPUT

Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
- Monolithic: create a new recommender model
- Mixed



HYBRID OUTPUT

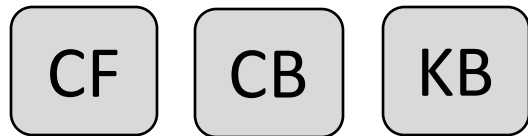


HYBRID OUTPUT

Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
- Monolithic: create a new recommender model
- Mixed: present predictions of different recommender models side by side



HYBRID OUTPUT

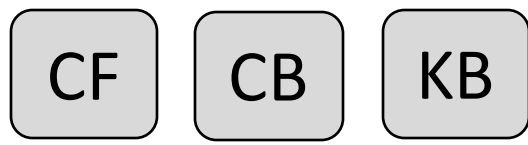


HYBRID OUTPUT

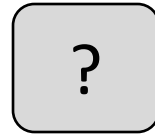
Hybrid recommendation

Three main types

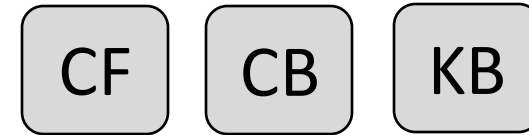
- Ensemble: combine predictions of different recommender models
- Monolithic: create a new recommender model
- Mixed: present predictions of different recommender models side by side



HYBRID OUTPUT



HYBRID OUTPUT



OUTPUT OF CF

OUTPUT OF CB

OUTPUT OF KB

Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
 - Combination of model outputs (parallel)
 - Pipelining (sequential)
- Monolithic: create a new recommender model
- Mixed: present predictions of different recommender models side by side

Combination of model outputs

Model A

Model B

Combination of model outputs

Collaborative Filtering

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?



User Rating Matrix

Model B

Combination of model outputs

Collaborative Filtering

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?



User Rating Matrix

Item Content Matrix



	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

Combination of model outputs



User Rating Matrix

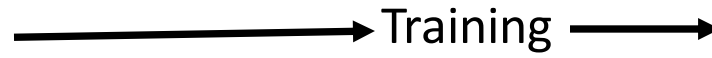
Item Content Matrix

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering



Model
A

Recommendation

Output
 R_A

User
Profile

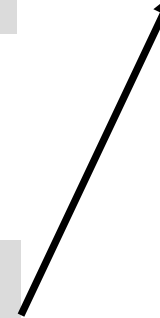
User Rating Matrix

Item Content Matrix

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
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Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

Combination of model outputs



	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

Training

Combination of model outputs

Model
A

Recommendation

Output
 R_A

User
Profile

User Rating Matrix

Item Content Matrix

Training

Model
B

Content Based

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

Training

Combination of model outputs

Model A

Recommendation

Output R_A

User Profile

Model B

Recommendation

Output R_B

Item Content Matrix

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

Training

Training

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

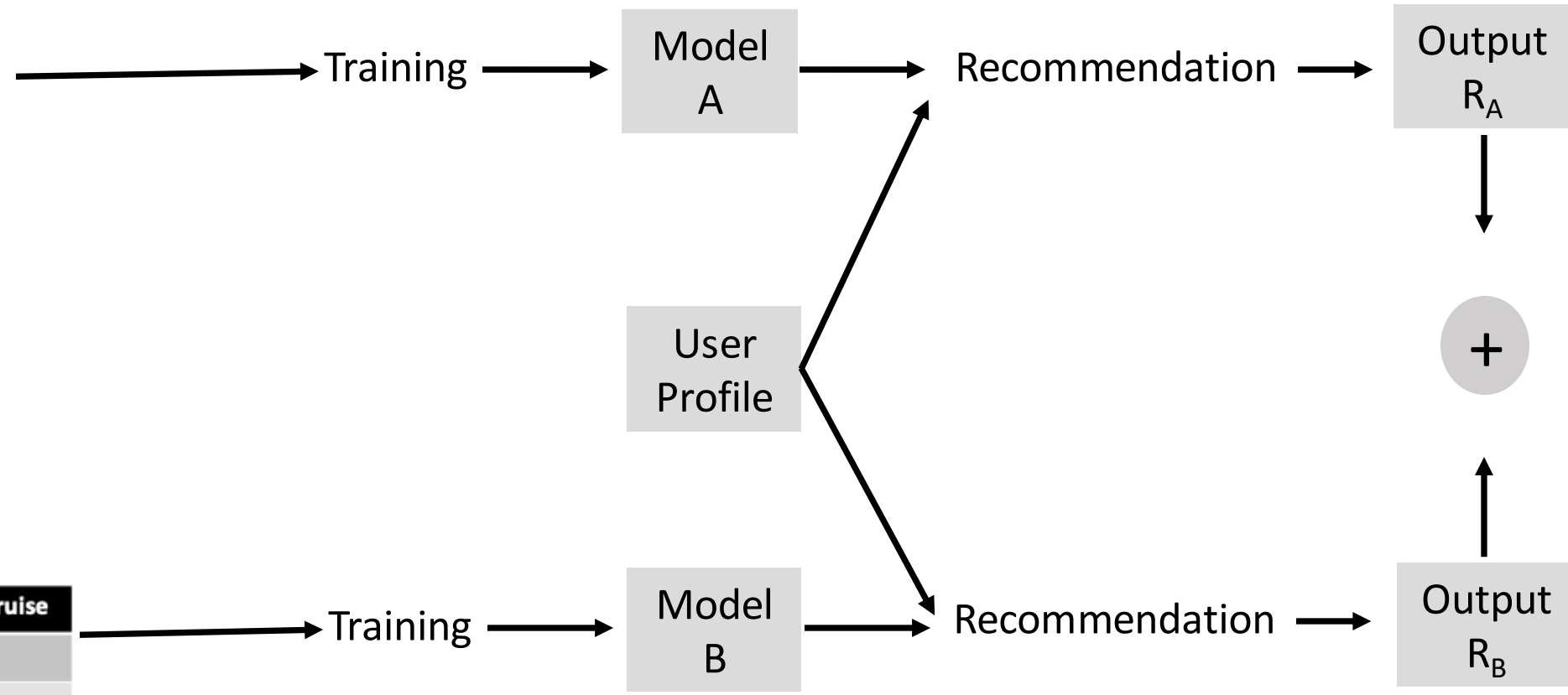
User Rating Matrix

Item Content Matrix

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

Combination of model outputs

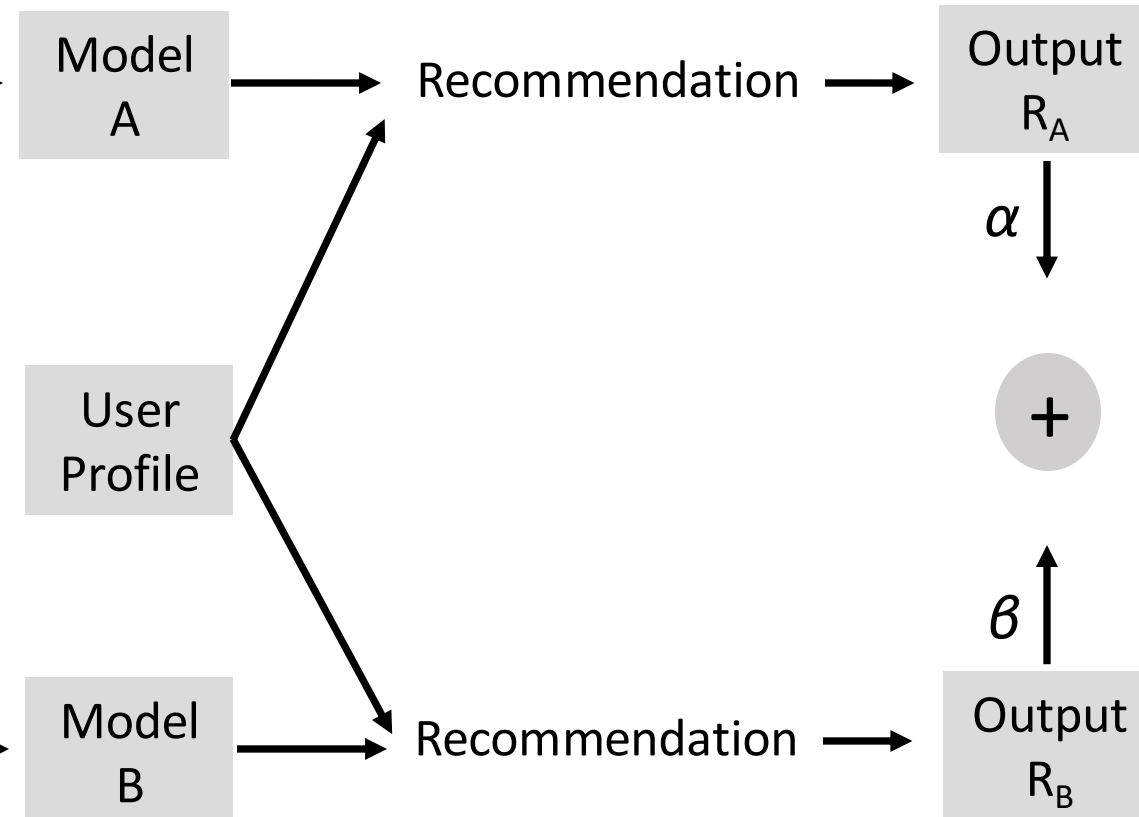


	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

Training

Combination of model outputs



$$R_H = \alpha R_A + \beta R_B$$

Linear (weighted sum)
Tune α , β with grid search

Item Content Matrix

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

Linear combination of model outputs

- Each model is used off-the-shelf

Linear combination of model outputs

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- Combine only outputs
- Models can have different inputs

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Advantages

- Easy
- Parallel execution

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Disadvantages

- Optimal values of weights depend on dataset

Linear combination of model outputs

- Each model is used off-the-shelf
- Combine only outputs
- Models can have different inputs

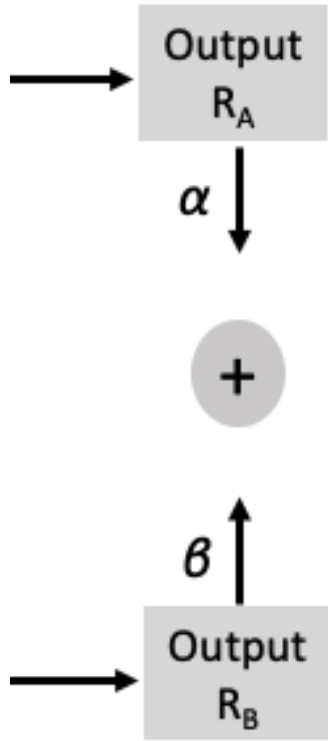
Advantages

- Easy
- Parallel execution

Disadvantages

- Optimal values of weights depend on dataset
- Works better when outputs are on the same scale

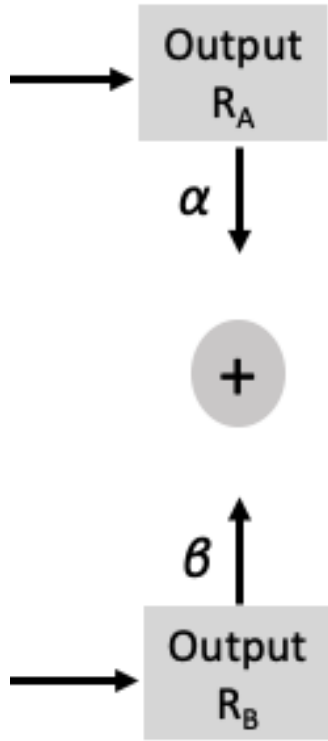
Linear combination of model outputs



Weighted sum

	R_A	R_B
MOVIE1	5	4
MOVIE3	4	3
MOVIE2	2	3
MOVIE5	1	1
MOVIE4	1	2

Linear combination of model outputs

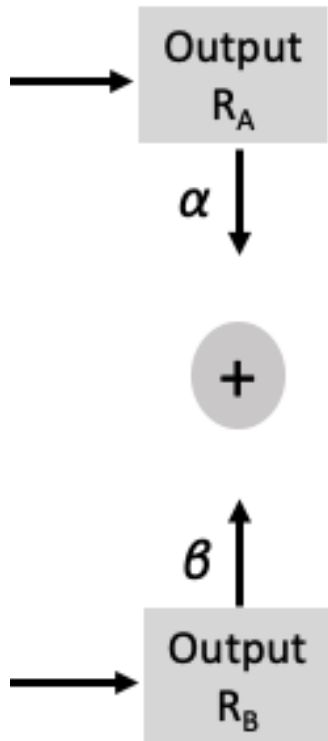


Weighted sum

	R_A	R_B	R_H
MOVIE1	$5 \alpha +$	4β	$= 4.6$
MOVIE3	$4 \alpha +$	3β	$= 3.6$
MOVIE2	$2 \alpha +$	3β	$= 2.4$
MOVIE5	$1 \alpha +$	1β	$= 1.0$
MOVIE4	$1 \alpha +$	2β	$= 1.4$

$$\alpha=0.6, \beta=0.4$$

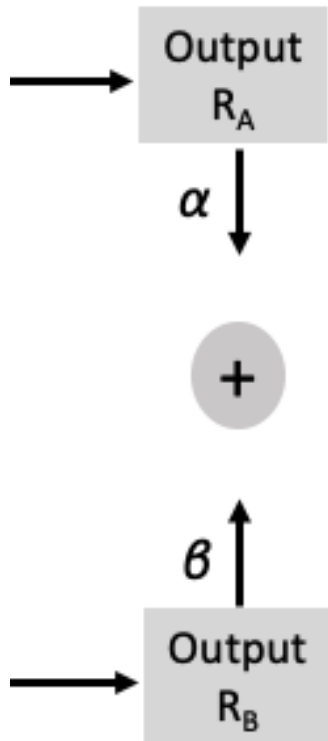
Combination of model outputs



CombMAX

	R_A	R_B
MOVIE1	5	4
MOVIE3	4	3
MOVIE2	2	3
MOVIE5	1	1
MOVIE4	1	2

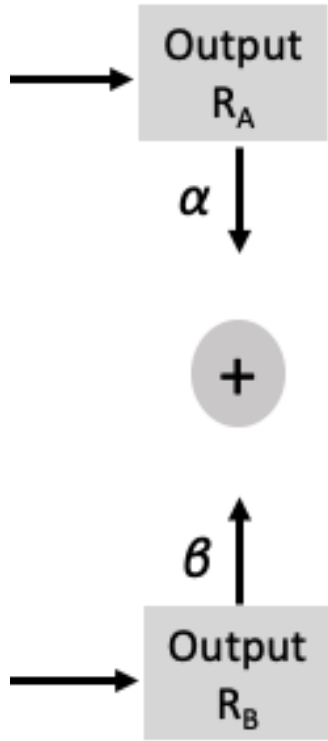
Combination of model outputs



CombMAX

	R_A	R_B	$R_{H_CombMAX}$
MOVIE1	5	4	= 5
MOVIE3	4	3	= 4
MOVIE2	2	3	= 3
MOVIE5	1	1	= 1
MOVIE4	1	2	= 2

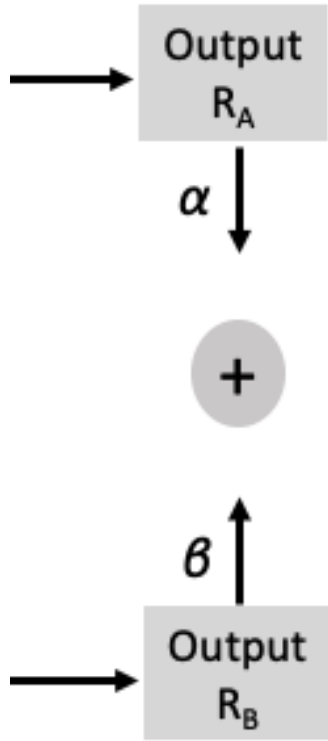
Combination of model outputs



CombMAX & CombMIN

	R_A	R_B	$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	= 5	= 4
MOVIE3	4	3	= 4	= 3
MOVIE2	2	3	= 3	= 2
MOVIE5	1	1	= 1	= 1
MOVIE4	1	2	= 2	= 1

Combination of model outputs

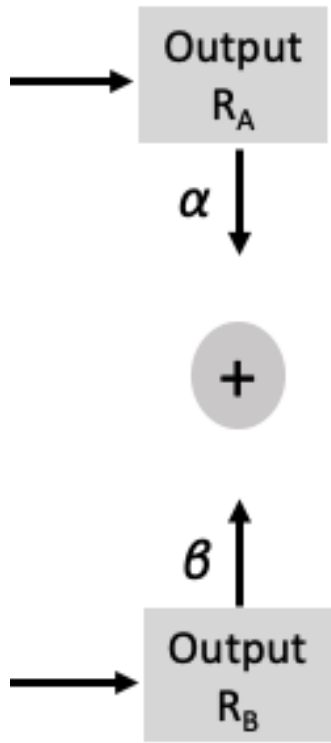


CombMAX & CombMIN

	R_A	R_B	$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	= 5	= 4
MOVIE3	4	3	= 4	= 3
MOVIE2	2	3	= 3	= 2
MOVIE5	1	1	= 1	= 1
MOVIE4	1	2	= 2	= 1

tie

Combination of model outputs



CombMAX & CombMIN

	R_A	R_B	$R_{H_CombMAX}$	$R_{H_CombMIN}$	
MOVIE1	5	4	= 5	= 4	Same ranking
MOVIE3	4	3	= 4	= 3	
MOVIE2	2	3	= 3	= 2	
MOVIE5	1	1	= 1	= 1	
MOVIE4	1	2	= 2	= 1	

Combination of model outputs

CombMAX & CombMIN

$R_{H_WeightedSum}$

= 4.6

MOVIE1

= 3.6

MOVIE3

= 2.4

MOVIE2

= 1.0

MOVIE5

= 1.4

MOVIE4

R_A

5

4

2

1

1

R_B

4

3

3

1

2

$R_{H_CombMAX}$

= 5

= 4

= 3

= 1

= 2

$R_{H_CombMIN}$

= 4

= 3

= 2

= 1

= 1

Same ranking

Combination of model outputs

	R_A	R_B	R_C
MOVIE1	5	4	3
MOVIE3	4	3	-
MOVIE2	2	-	-
MOVIE5	1	-	2
MOVIE4	1	2	3
MOVIE6	-	5	5

Combination of model outputs

	R_A	R_B	R_C	$R_{H_CombMAX}$
MOVIE1	5	4	3	= 5
MOVIE3	4	3	-	= 4
MOVIE2	2	-	-	= 2
MOVIE5	1	-	2	= 2
MOVIE4	1	2	3	= 3
MOVIE6	-	5	5	= 5

Combination of model outputs

	R_A	R_B	R_C		$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	3		= 5	= 3
MOVIE3	4	3	-		= 4	= 3
MOVIE2	2	-	-		= 2	= 2
MOVIE5	1	-	2		= 2	= 1
MOVIE4	1	2	3		= 3	= 1
MOVIE6	-	5	5		= 5	= 5

Combination of model outputs

	R_A	R_B	R_C	$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	3	= 5	= 3
MOVIE3	4	3	-	= 4	= 3
MOVIE2	2	-	-	= 2	= 2
MOVIE5	1	-	2	= 2	= 1
MOVIE4	1	2	3	= 3	= 1
MOVIE6	-	5	5	= 5	= 5

ties

Combination of model outputs

CombMNZ

	R_A	R_B	R_C	$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	3	= 5	= 3
MOVIE3	4	3	-	= 4	= 3
MOVIE2	2	-	-	= 2	= 2
MOVIE5	1	-	2	= 2	= 1
MOVIE4	1	2	3	= 3	= 1
MOVIE6	-	5	5	= 5	= 5

Sum of ratings per item x number of lists where item occurs

Combination of model outputs

CombMNZ

	R_A	R_B	R_C	$R_{H_CombMNZ}$	$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	3	= 5+4+3	= 5	= 3
MOVIE3	4	3	-	= 4+3	= 4	= 3
MOVIE2	2	-	-	= 2	= 2	= 2
MOVIE5	1	-	2	= 1+2	= 2	= 1
MOVIE4	1	2	3	= 1+2+3	= 3	= 1
MOVIE6	-	5	5	= 5+5	= 5	= 5

Sum of ratings per item x number of lists where item occurs

Combination of model outputs

CombMNZ

	R_A	R_B	R_C	$R_{H_CombMNZ}$	$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	3	$= (5+4+3)3$	$= 5$	$= 3$
MOVIE3	4	3	-	$= (4+3)2$	$= 4$	$= 3$
MOVIE2	2	-	-	$= (2)1$	$= 2$	$= 2$
MOVIE5	1	-	2	$= (1+2)2$	$= 2$	$= 1$
MOVIE4	1	2	3	$= (1+2+3)3$	$= 3$	$= 1$
MOVIE6	-	5	5	$= (5+5)2$	$= 5$	$= 5$

Sum of ratings per item x **number of lists where item occurs**

Combination of model outputs

CombMNZ

	R_A	R_B	R_C	$R_{H_CombMNZ}$		$R_{H_CombMAX}$	$R_{H_CombMIN}$
MOVIE1	5	4	3	$= (5+4+3)3$	$= 36$	$= 5$	$= 3$
MOVIE3	4	3	-	$= (4+3)2$	$= 14$	$= 4$	$= 3$
MOVIE2	2	-	-	$= (2)1$	$= 2$	$= 2$	$= 2$
MOVIE5	1	-	2	$= (1+2)2$	$= 6$	$= 2$	$= 1$
MOVIE4	1	2	3	$= (1+2+3)3$	$= 18$	$= 3$	$= 1$
MOVIE6	-	5	5	$= (5+5)2$	$= 20$	$= 5$	$= 5$

Sum of ratings per item x number of lists where item occurs

Combination of model outputs

- Weighted Sum
- CombMAX
- CombMIN
- CombMNZ

Combination of model outputs

Combinations of ratings (e.g. 0 to 5)

- Weighted Sum
- CombMAX
- CombMIN
- CombMNZ

	RATINGS
MOVIE1	5
MOVIE3	5
MOVIE2	4
MOVIE5	4
MOVIE4	4
MOVIE9	3
MOVIE8	2
MOVIE6	2
MOVIE11	1
MOVIE7	0

Combination of model outputs

Combinations of ratings (e.g. 0 to 5)

- Weighted Sum
- CombMAX
- CombMIN
- CombMNZ

Combinations of rank positions

	RATINGS	RANK POSITION
MOVIE1	5	1
MOVIE3	5	2
MOVIE2	4	3
MOVIE5	4	4
MOVIE4	4	5
MOVIE9	3	6
MOVIE8	2	7
MOVIE6	2	8
MOVIE11	1	9
MOVIE7	0	10

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid
Movie1

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid
Movie1
Movie5

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid
Movie1
Movie5
Movie2

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid
Movie1
Movie5
Movie2
Movie6

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid
Movie1
Movie5
Movie2
Movie6

Round Robin



Source: <https://worldbirds.com/robin-symbolism/>

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie5
Movie2	Movie6
Movie3	Movie7
Movie4	Movie8



Hybrid
Movie1
Movie5
Movie2
Movie6

Round Robin



Source: <https://worldbirds.com/robin-symbolism/>



Source: <https://rosemille.com/products/double-faced-silk-satin-ribbon?variant=508388533>

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid

Round Robin

Combination of model outputs (rank positions)

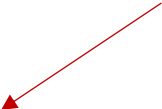
Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid

Round Robin

Which one is correct?



Movie1	Movie1	Movie1
Movie5	Movie2	Movie2
Movie2	Movie5	Movie5
Movie4	Movie4	Movie3

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid

Round Robin

Movie1	Movie1	Movie1
Movie5	Movie2	Movie2
Movie2	Movie5	Movie5
Movie4	Movie4	Movie3

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1

Round Robin

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1

Round Robin

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1
Movie2

Round Robin

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1
Movie2
Movie5

Round Robin

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1
Movie2
Movie5
Movie3

Round Robin

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1
Movie2
Movie5
Movie3

Round Robin

Advantage: independent of output scale

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie5
Movie3	Movie4
Movie4	Movie6



Hybrid
Movie1
Movie2
Movie5
Movie3

Round Robin

Advantage: independent of output scale

Disadvantage: hard to calibrate the quality of each model

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	
Movie2	Movie3	
Movie3	Movie5	
Movie4	Movie4	

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	
Movie2	Movie3	
Movie3	Movie5	
Movie4	Movie4	0

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	
Movie2	Movie3	
Movie3	Movie5	1
Movie4	Movie4	0

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

For n rank positions,
-> bottom rank has 0 points
-> top rank has $n-1$ points

Borda Count
(or BordaFuse)



Source: https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Movie1: 6 points

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Movie1: 6 points

Movie2: 2 points

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Movie1: 6 points

Movie2: 2 points

Movie3: 3 points

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Movie1: 6 points

Movie2: 2 points

Movie3: 3 points

Movie4: 0 points

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Movie1: 6 points

Movie2: 2 points

Movie3: 3 points

Movie4: 0 points

Movie5: 1 point

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0



Hybrid
Movie1
Movie3
Movie2
Movie5

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Movie1: 6 points
Movie2: 2 points
Movie3: 3 points
Movie4: 0 points
Movie5: 1 point

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0



Hybrid
Movie1
Movie3
Movie2
Movie5

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Advantage: considers importance of rank position

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0



Hybrid
Movie1
Movie3
Movie2
Movie5

Borda Count
(or BordaFuse)



Source:https://en.wikipedia.org/wiki/Jean-Charles_de_Borda

Advantage: considers importance of rank position

Disadvantage: assumes “equal distance” between ranks

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Borda Count
Movie1
Movie3
Movie2
Movie5

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Round Robin	Borda Count
Movie1	Movie1
Movie2	Movie3
Movie3	Movie2
Movie5	Movie5

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Round Robin	Borda Count
Movie1	Movie1
Movie2	Movie3
Movie3	Movie2
Movie5	Movie5

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Round Robin	Borda Count
Movie1	Movie1
Movie2	Movie3
Movie3	Movie2
Movie5	Movie5

Which one is better?

Combination of model outputs (rank positions)

Model A	Model B	points
Movie1	Movie1	3
Movie2	Movie3	2
Movie3	Movie5	1
Movie4	Movie4	0

Round Robin	Borda Count
Movie1	Movie1
Movie2	Movie3
Movie3	Movie2
Movie5	Movie5

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Round Robin	Borda Count

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Round Robin	Borda Count
Movie1	Movie1
Movie2	Movie3
Movie3	Movie2
Movie4	Movie4

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Round Robin	Borda Count
Movie1	Movie1
Movie2	Movie3
Movie3	Movie2
Movie4	Movie4

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-				
M2		-			
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2			
M2		-			
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3		
M2		-			
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	
M2		-			
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2		-			
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-			
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1		
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3			-		
M4				-	
M5					-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method



Source:https://en.wikipedia.org/wiki/Marquis_de_Condorcet

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Sum of wins
M1 wins:
M2 wins:
M3 wins:
M4 wins:
M5 wins:

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Sum of wins
M1 wins: 9
M2 wins:
M3 wins:
M4 wins:
M5 wins:

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Sum of wins
M1 wins: 9
M2 wins: 3
M3 wins:
M4 wins:
M5 wins:

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Sum of wins
M1 wins: 9
M2 wins: 3
M3 wins: 5
M4 wins:
M5 wins:

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid

Condorcet method

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Sum of wins
M1 wins: 9
M2 wins: 3
M3 wins: 5
M4 wins: 1
M5 wins: 0

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4



Hybrid
Movie1
Movie3
Movie2
Movie4

Condorcet method

	M1	M2	M3	M4	M5
M1	-	2	3	3	1
M2	0	-	1	2	0
M3	0	1	-	3	1
M4	0	0	0	-	1
M5	0	0	0	0	-

Pairwise comparisons

Sum of wins
M1 wins: 9
M2 wins: 3
M3 wins: 5
M4 wins: 1
M5 wins: 0

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Condorcet
Movie1
Movie3
Movie2
Movie4

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Condorcet	Round Robin	Borda Count
Movie1	Movie1	Movie1
Movie3	Movie2	Movie3
Movie2	Movie3	Movie2
Movie4	Movie4	Movie4

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Condorcet	Round Robin	Borda Count
Movie1	Movie1	Movie1
Movie3	Movie2	Movie3
Movie2	Movie3	Movie2
Movie4	Movie4	Movie4

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Condorcet	Round Robin	Borda Count
Movie1	Movie1	Movie1
Movie3	Movie2	Movie3
Movie2	Movie3	Movie2
Movie4	Movie4	Movie4

Combination of model outputs (rank positions)

Model A	Model B	Model C
Movie1	Movie1	Movie1
Movie2	Movie3	Movie3
Movie3	Movie4	Movie2
Movie4	Movie5	Movie4

Condorcet	Round Robin	Borda Count
Movie1	Movie1	Movie1
Movie3	Movie2	Movie3
Movie2	Movie3	Movie2
Movie4	Movie4	Movie4

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m)$$

m: movie

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{r_j(m)}$$

m : movie
 r_j : rank position

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

let $k = 0$

Movie1:

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

let $k = 0$

Movie1: $1/1 + 1/1 = 2$

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$

Movie2: $1/2 + 0 = 0.5$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$

Movie2: $1/2 + 0 = 0.5$

Movie3: $1/3 + 1/2 = 0.83$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$

Movie2: $1/2 + 0 = 0.5$

Movie3: $1/3 + 1/2 = 0.83$

Movie4: $1/4 + 1/4 = 0.5$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$

Movie2: $1/2 + 0 = 0.5$

Movie3: $1/3 + 1/2 = 0.83$

Movie4: $1/4 + 1/4 = 0.5$

Movie5: $0 + 1/3 = 0.33$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1$	$= 2$	
Movie2: $1/2 + 0$	$= 0.5$	
Movie3: $1/3 + 1/2$	$= 0.83$	tie
Movie4: $1/4 + 1/4$	$= 0.5$	
Movie5: $0 + 1/3$	$= 0.33$	

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$
Movie2: $1/2 + 0 = 0.5$
Movie3: $1/3 + 1/2 = 0.83$
Movie4: $1/4 + 1/4 = 0.5$
Movie5: $0 + 1/3 = 0.33$

let $k = 60$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$
Movie2: $1/2 + 0 = 0.5$
Movie3: $1/3 + 1/2 = 0.83$
Movie4: $1/4 + 1/4 = 0.5$
Movie5: $0 + 1/3 = 0.33$

let $k = 60$

Movie1: $1/61 + 1/61 =$
Movie2: $1/62 + 0 =$
Movie3: $1/63 + 1/62 =$
Movie4: $1/64 + 1/64 =$
Movie5: $0 + 1/63 =$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Reciprocal Rank Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$
Movie2: $1/2 + 0 = 0.5$
Movie3: $1/3 + 1/2 = 0.83$
Movie4: $1/4 + 1/4 = 0.5$
Movie5: $0 + 1/3 = 0.33$

let $k = 60$

Movie1: $1/61 + 1/61 = 0.033$
Movie2: $1/62 + 0 = 0.016$
Movie3: $1/63 + 1/62 = 0.032$
Movie4: $1/64 + 1/64 = 0.031$
Movie5: $0 + 1/63 = 0.015$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Hybrid
Movie1
Movie3
Movie4
Movie2

Reciprocal Rank
Fusion (RRF)

$$RRF(m) = \sum_j \frac{1}{k + r_j(m)}$$

m : movie
 r_j : rank position
 k : parameter

let $k = 0$

Movie1: $1/1 + 1/1 = 2$
Movie2: $1/2 + 0 = 0.5$
Movie3: $1/3 + 1/2 = 0.83$
Movie4: $1/4 + 1/4 = 0.5$
Movie5: $0 + 1/3 = 0.33$

let $k = 60$

Movie1: $1/61 + 1/61 = 0.033$
Movie2: $1/62 + 0 = 0.016$
Movie3: $1/63 + 1/62 = 0.032$
Movie4: $1/64 + 1/64 = 0.031$
Movie5: $0 + 1/63 = 0.015$

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Rec. Rank Fusion (<i>k</i> =60)	Round Robin	Borda Count	Condorcet
Movie1			
Movie3			
Movie4			
Movie2			

Combination of model outputs (rank positions)

Model A	Model B
Movie1	Movie1
Movie2	Movie3
Movie3	Movie5
Movie4	Movie4

Rec. Rank Fusion (<i>k</i> =60)	Round Robin	Borda Count	Condorcet
Movie1	Movie1	Movie1	Movie1
Movie3	Movie2	Movie3	Movie3
Movie4	Movie3	Movie2	Movie2
Movie2	Movie5	Movie5	Movie5

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Movie4	Movie3	Movie2	Movie2
Movie2	Movie5	Movie5	Movie5

Reciprocal Rank Fusion rewards items close the top & penalises items close the bottom

Resolving ties

Resolving ties

- Use the average of the ranks that the tied elements cover

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- Use the highest rank for all tied elements

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Resolving ties

- Use the average of the ranks that the tied elements cover
 - Use the highest rank for all tied elements
 - And even more complex methods...
-
- Toss a coin
 - Use lexicographic order

QUIZ WS3

- Open a browser
- Go to: www.socrative.com
- Login as a student
- Under Room Name, type: LIOMA
- Start doing the quiz



Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
 - Combination of model outputs (parallel)
 - **Pipelining (sequential)**
- Monolithic: create a new recommender model
- Mixed: present predictions of different recommender models side by side

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Content Based

Pipelining (sequential) model combination

Pipelining (sequential) model combination

Content Based

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

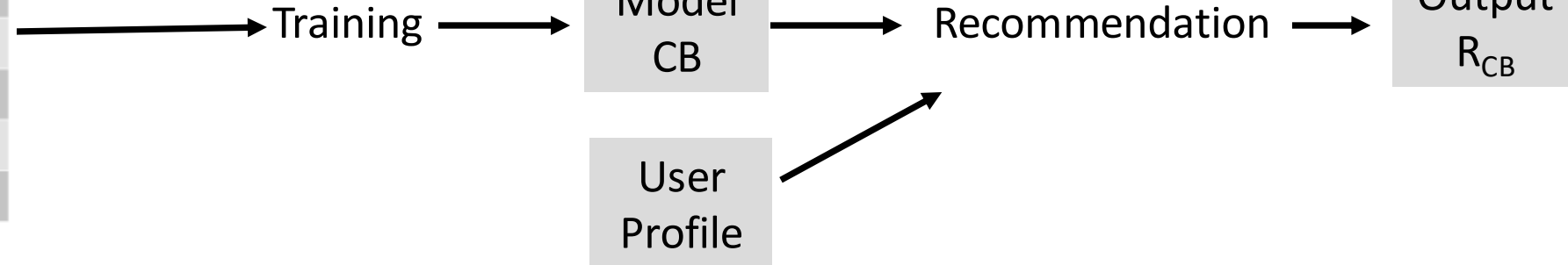
→ Training →

Model
CB

Pipelining (sequential) model combination

Content Based

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0



Pipelining (sequential) model combination

Content Based

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Training

Model
CB

Recommendation

Output
 R_{CB}

User
Profile

Model
CF

Training

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

Pipelining (sequential) model combination

Content Based

	action	romance	Tom Cruise
Movie1	0.9	0.1	1
Movie2	0.7	0	1
Movie3	0.8	0.2	0
Movie4	0.7	0	0
Movie5	0.2	0.9	0
Movie6	0.1	0.9	0
Movie7	0.2	0.9	0

Training

Model
CB

Recommendation

Output
 R_{CB}

User
Profile

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Collaborative Filtering

Output
 R_H

Recommendation

Model
CF

Training

User
Profile

Pipelining (sequential) model combination

- Chain models so that the output of one model is fed as input to another model

Pipelining (sequential) model combination

- Chain models so that the output of one model is fed as input to another model

Advantage

- Increased density of User Rating Matrix gives better recommendations

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Pipelining (sequential) model combination

- Chain models so that the output of one model is fed as input to another model

Advantage

- Increased density of User Rating Matrix gives better recommendations

Disadvantages

- Computational efficiency (memory, runtime, ...)

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Pipelining (sequential) model combination

- Chain models so that the output of one model is fed as input to another model

Advantage

- Increased density of User Rating Matrix gives better recommendations

Disadvantages

- Computational efficiency (memory, runtime, ...)
- Too dense User Rating Matrix can lead to overfitting

	User1	User2	User3	User4
Movie1	5	5	0	0
Movie2	5	?	?	0
Movie3	?	4	0	?
Movie4	4	5	0	1
Movie5	0	1	5	4
Movie6	0	0	5	4
Movie7	0	0	5	?

Hybrid recommendation

Three main types

- Ensemble: combine predictions of different recommender models
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HYBRID OUTPUT

Monolithic (merge model components)

- Merge model components, if models have compatible structure

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- Merge model components, if models have compatible structure
 - Matrices

$$\text{Matrix}_{\text{Hybrid}} = \alpha \text{Matrix}_{\text{ModelA}} + (1-\alpha)\text{Matrix}_{\text{ModelB}}$$

- Tune α with grid search

Recap

Types of hybrid recommendation

- Ensemble
 - Parallel combination of outputs
 - Combinations of ratings: weighted sum, combMAX, combMIN, combMZ
 - Combinations of rank positions: round robin, Borda, Condorcet, RRF
 - Sequential combination
- Monolithic
- Mixed

QUIZ WS4

- Open a browser
- Go to: www.socrative.com
- Login as a student
- Under Room Name, type: LIOMA
- Start doing the quiz

